

Coordinate-invariant routing coordinates for multi-physics machine learning: what the geometry of the operating manifold does, and does not, buy

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Abstract

Machines trace trajectories through coupled thermal, mechanical and electrical state spaces, and one global network must serve every regime along the way. A natural proposal is to route computation on the local geometry of that space. We formalise the proposal and test it. For a diffusion the canonical metric is $g = V^{-1}$, the inverse diffusion tensor, so volatility is not a feature sitting alongside curvature: it *is* the geometry. The routing coordinates then reduce to two scalars, the scalar curvature R and a drift-to-noise Péclet number Pe , both invariant under smooth relabelling of the sensors. We prove an exact identity governing how sharply an operator assembly may switch and verify it to 2×10^{-17} . Across five systems — a 1.3 M sample permanent-magnet motor bench, NASA C-MAPSS, and three MuJoCo Menagerie arms carrying injected joint thermal networks — geometric routing does *not* improve in-distribution accuracy over routing on the raw state at matched capacity. Pe survives a coordinate change almost exactly ($\rho = 0.999$) while the curvature does not ($\rho = 0.60$), and where increments are a jump process rather than a diffusion the metric is misspecified and geometric routing is actively worse. We report where the construction earns its cost and where it does not.

1 Introduction

A machine under load does not sit at an operating point; it moves along one. A servo joint accelerating a payload dissipates ohmic heat into a winding whose rising temperature derates the torque available to the next command, on a thermal time constant three decades slower than the control loop that issued it. A traction motor’s rotor temperature, which no production sensor can reach, is a functional of the whole recent history of current and speed. In both cases the quantity to be predicted depends on where the machine is on a coupled multi-physics trajectory, and a single monolithic network must serve every part of that trajectory at once.

Mixture-of-experts routing is the standard answer, and it has been applied to exactly these problems: multi-gate mixtures for turbofan remaining-useful-life, fault-mode gating, regime-conditioned prognostics. In all of these the router reads the raw state. A second, separate literature routes on *geometry*, but in graph representation learning, where discrete Ollivier–Ricci curvature selects among embedding spaces of different constant curvature. Nothing yet routes physical computation on the intrinsic geometry of an estimated continuous state manifold.

That is the proposal we formalise and test. The intuition is appealing: regions where trajectories separate quickly, or where noise dominates drift, are physically distinct and might deserve distinct computation. The intuition also makes a sharper and more testable promise than accuracy. Curvature is a coordinate invariant. If routing decisions are functions of invariants, they are

unchanged when the sensors are relabelled — recalibrated, rescaled, nonlinearly warped — whereas a router reading the raw state is not. Fleets are recalibrated constantly. This is the claim worth testing, and it is the one we test.

Our contributions are as follows.

1. **The volatility is the metric.** Treating curvature and volatility as two features to concatenate double-counts. For a non-degenerate diffusion the canonical metric is $g = V^{-1}$, in which Varadhan’s short-time heat-kernel asymptotics give geodesic distance. The routing coordinates collapse to two scalars, and become $\sim 100\times$ cheaper to compute (Section 2).
2. **An exact switching identity.** The sensitivity of an assembly to its routing coordinate is exactly a gate-weighted covariance between router logit gradients and expert outputs (Theorem 1); verified to 2×10^{-17} .
3. **A negative in-distribution result, with controls.** At matched parameter count, matched expert bank and matched expert inputs — so that only the router’s input differs — geometric routing does not beat raw-state routing on any of five systems. Controls for dimensionality, for basis dependence and for cheap non-geometric proxies are all run.
4. **A diagnosis.** P_e is estimable and invariant; R is theoretically the more attractive coordinate and empirically the less reliable one, because it is a second derivative of an estimated diffusion field. Where increments are jumps rather than diffusion, the whole construction is misspecified and we show it failing.

2 The construction

2.1 State manifold and chart

Let the operating state be $\mathbf{s}(t) \in \mathcal{M} \subset \mathbb{R}^d$, collecting the measured thermal, mechanical and electrical channels. A chart $\phi : \mathcal{M} \rightarrow \mathbb{R}^k$ with $k \ll d$ carries the ambient sensor vector to local manifold coordinates $\mathbf{z} = \phi(\mathbf{s})$; we use the encoder of an autoencoder fitted to the state trajectories.

The chart is not cosmetic. The scalar curvature requires the Riemann tensor, $O(k^4)$ contractions, and third derivatives of the network carrying the metric. Measured cost rises from 1.2 ms per point at $k = 2$ to 115 ms at $k = 8$ (Fig. 1a). Geometry is affordable only on a low-dimensional chart, which is what a chart is for. Smooth activations are mandatory throughout: with ReLU the curvature field is identically zero.

2.2 The volatility is the metric

Model the chart coordinates as a Stratonovich diffusion

$$d\mathbf{z} = \mathbf{a}(\mathbf{z}) dt + \Sigma(\mathbf{z}) \circ d\mathbf{W}, \quad V = \Sigma \Sigma^\top, \quad (1)$$

whose drift and diffusion are the first two conditional Kramers–Moyal moments. We fit them by exact heteroscedastic Gaussian likelihood on one-step transitions, which is the maximum-likelihood form of the Euler–Maruyama transition density, with Σ carried as a Cholesky factor.

It is tempting to hand a router the state alongside separate curvature and volatility features, as $[\mathbf{s} \parallel R \parallel \text{Tr } V]$. This double-counts. For a non-degenerate diffusion the canonical Riemannian structure on the state space is

$$g(\mathbf{z}) = V(\mathbf{z})^{-1}, \quad (2)$$

the metric in which Varadhan’s short-time asymptotics reduce to geodesic distance, $-4t \log p(\mathbf{z}, \mathbf{z}', t) \rightarrow d_g(\mathbf{z}, \mathbf{z}')^2$. The volatility field is the geometry. Curvature and volatility are one object, and every routing coordinate below is a scalar built from g . Because g comes directly from the fitted Cholesky factor, the curvature needs only second derivatives of one small network rather than third derivatives of a multi-step flow map.

2.3 Two invariants

Definition 1 (Routing coordinates).

$$R(\mathbf{z}) = g^{jk} \text{Ric}_{jk}, \quad \text{Pe}(\mathbf{z}) = \|\mathbf{a}\|_g^2 = \mathbf{a}^\top V^{-1} \mathbf{a}. \quad (3)$$

R measures how the volatility geometry bends. Pe is a local Péclet number, a squared drift-to-noise ratio distinguishing advective operating points ($\text{Pe} \gg 1$) from diffusive ones ($\text{Pe} \ll 1$). Both are dimensionless in the standardised chart coordinates we work in; Pe carries dimensions of inverse time in general and is reported against the telemetry interval.

Proposition 1 (Invariance). *Let f be a diffeomorphism of the state coordinates. Under the Stratonovich convention $\mathbf{a}' = J_f \mathbf{a}$ and $V' = J_f V J_f^\top$, so $g' = J_f^{-\top} g J_f^{-1}$ and*

$$\text{Pe}' = \mathbf{a}'^\top J_f^\top J_f^{-\top} V^{-1} J_f^{-1} J_f \mathbf{a} = \text{Pe}, \quad (4)$$

while R is a scalar invariant by construction.

The Stratonovich convention is load-bearing. Under Itô the drift acquires a correction under change of variables and Pe would be only approximately invariant; the fitted Euler–Maruyama drift is an Itô drift and is converted by $a^i = b^i - \frac{1}{2} \Sigma^{jk} \partial_j \Sigma^{ik}$. We verify Proposition 1 against an analytic push-forward to relative error 1.9×10^{-8} for Pe and 9.7×10^{-7} for R (Fig. 2b).

2.4 How sharply may an assembly switch?

Write the assembly $F(\mathbf{x}; \mathbf{r}) = \sum_k p_k(\mathbf{r}) \mathcal{O}_k(\mathbf{x})$ with $p = \text{softmax}(u(\mathbf{r})/\tau)$.

Theorem 1 (Routing sensitivity).

$$\frac{\partial F}{\partial \mathbf{r}} = \frac{1}{\tau} \text{Cov}_{k \sim p} \left(\frac{\partial u_k}{\partial \mathbf{r}}, \mathcal{O}_k(\mathbf{x}) \right), \quad (5)$$

hence $\|\partial F / \partial \mathbf{r}\| \leq \tau^{-1} \text{sd}_p(\partial u / \partial \mathbf{r}) \text{sd}_p(\mathcal{O})$.

Proof. $\partial_{\mathbf{r}} F = \sum_k (\partial_{\mathbf{r}} p_k) \mathcal{O}_k$ and $\partial_{\mathbf{r}} p_k = \tau^{-1} p_k (\partial_{\mathbf{r}} u_k - \mathbb{E}_p[\partial_{\mathbf{r}} u])$. Substituting gives $\tau^{-1} \sum_k p_k (\partial_{\mathbf{r}} u_k - \mathbb{E}_p[\partial_{\mathbf{r}} u]) \mathcal{O}_k$, which is the stated covariance because the mean-zero first factor annihilates any constant shift of $\mathcal{O}$. Cauchy–Schwarz gives the bound. $\square$

Observing that a softmax of smooth experts is smooth is true and empty. Equation (5) is an identity, not a bound, and says what actually controls the size of a transition: the assembly moves only where the experts *disagree* and the gate is moving at the same time. Smooth handover across an operating boundary is therefore bought either with gate temperature or with expert agreement in the overlap, and the two are interchangeable at fixed product. We verify (5) to a maximum residual of 2.1×10^{-17} across gate temperatures (Fig. 1c).

3 Systems

Five systems span two physics families and both real and simulated telemetry (Table 3).

PMSM motor bench (real). 1,330,816 samples at 2 Hz from a 52 kW permanent-magnet synchronous machine, 69 independent measurement sessions. The task is the real industrial one: predict the four internal temperatures — rotor magnet, stator yoke, tooth and winding — from the operating condition alone, since no production machine instruments a spinning rotor. The manifold state is the eight-dimensional operating condition only (drive voltages and currents, speed, torque, ambient and coolant); the four temperatures being predicted are never part of it, so the geometric coordinates cannot leak the target. Splits are by session.

C-MAPSS FD004 (simulated, standard benchmark). NASA turbofan run-to-failure with six discrete flight regimes and two fault modes; the task is remaining useful life with the standard piecewise-linear cap at 125 cycles. We include it because its six regimes are the closest public analogue of a piecewise-charted operating manifold — and, as it turns out, because it breaks the diffusion assumption (Section 5).

Three MuJoCo Menagerie arms with injected thermals (simulated). Rigid-body dynamics come from Menagerie’s manufacturer-parameterised UR5e, Franka Panda and KUKA iiwa14. We add the physics Menagerie does not model: a two-node lumped-parameter thermal network per joint,

$$C_w \dot{T}_w = k_{cu} \tau^2 - (T_w - T_h)/R_{wh}, \quad (6)$$

$$C_h \dot{T}_h = (T_w - T_h)/R_{wh} - (T_h - T_{\text{env}})/R_{ha}, \quad (7)$$

with $\tau_w \approx 40$ s and $\tau_h \approx 18$ min against a 500 Hz control loop — a three-decade separation which is what makes the problem genuinely multi-physics. The coupling runs both ways: a hot winding is derated, and at $T_w > 110^\circ\text{C}$ the drive trips with hysteresis and a holding brake takes the load, producing a sharp physical switching boundary. Actuator noise grows with speed and with heat, so $V(\mathbf{s})$ is neither isotropic nor constant. The ohmic gain is fixed by motor physics, $k_{cu} = \Delta T_{\text{rated}}/[(R_{wh} + R_{ha})\tau_{\text{cont}}^2]$ with $\tau_{\text{cont}} = 0.35 \tau_{\text{peak}}$, not tuned to hit a target temperature. Excitation is band-limited multi-sine, the standard family in robot dynamic-parameter identification, with position and speed budgets shared across components so the commanded trajectory stays inside datasheet limits. Ground-truth regime labels are recorded and never shown to any model.

A consequence we report rather than engineer away: under identical excitation the platforms differ in thermal headroom. The Panda reaches all five regimes including drive trips, the UR5e four, and the iiwa14 — oversized for this duty at 0.20 of rated torque — stays within 2°C of ambient and never leaves the nominal regime.

4 Validation

Nothing downstream is meaningful if the geometry is wrong, so each layer is pinned separately.

Curvature. The scalar-curvature implementation reproduces closed forms to machine precision: the 2-sphere, the scaled Poincaré half-plane, the torus (non-constant curvature), a product $S^2 \times \mathbb{R}^2$ testing that flat directions contribute nothing, flat $\mathbb{R}^5$ (exactly zero), and invariance under a nonlinear reparameterisation. Worst error over all six: 1.3×10^{-15} .

Theorem 1. Maximum residual 2.1×10^{-17} over gate temperatures $\tau \in [0.25, 4]$, with the predicted $1/\tau$ scaling recovered exactly at fixed weights (Fig. 1b,c).

Invariance under estimation. This is the load-bearing test, and it splits the two coordinates apart. On an analytic pushed-forward system the algebra is exact for both. But when the SDE is

Table 1: In-distribution test error by router input. One expert bank, one budget, identical expert inputs; only the router input differs. Percentages and paired p -values are against the raw-state router over common seeds.

router input	C-MAPSS (cycles ²)	
	error	vs raw
monolith	299.3 ± 12.7	+0.5%
raw state	297.9 ± 3.7	—
invariants (R, Pe)	296.0 ± 2.8	-0.6%
raw + invariants	295.2 ± 5.3	-0.9%
$\text{Tr } V, \log \det V$	290.5 ± 1.3	-2.5%
local activity	301.7 ± 5.7	+1.3%
random projection	299.2 ± 2.9	+0.4%

refitted from scratch in the new coordinates from finite samples, Pe still agrees across the coordinate change at Spearman $\rho = 0.999$ and with ground truth at 0.998, whereas R agrees at only $\rho = 0.604 \pm 0.060$ across the change and 0.716 with truth (Fig. 2a). The curvature is a second derivative of an estimated diffusion field; finite telemetry does not pin it down. We regard this as the single most important empirical fact in the paper, and it is not the one the framework’s intuition predicts.

The basis-dependent control is instructive in the other direction: raw $\text{Tr } V$ transfers across this particular diffeomorphism at $\rho = 0.972$. Invariance is a guarantee, not an advantage, whenever the coordinate change at hand happens to be mild.

5 Results

5.1 In-distribution accuracy

Every arm shares one expert bank, one training budget and one set of expert inputs; only the router’s input differs. Alongside the raw-state router we run three controls: *random*, a fixed random two-dimensional projection, controlling for the mere fact of extra router inputs; *naive* ($\text{Tr } V, \log \det V$), the basis-dependent stochastic features, controlling for whether invariance is what matters; and *activity* (local speed and variance), a deliberately cheap non-geometric proxy controlling for whether anything local does just as well.

The result is a null. On C-MAPSS no arm differs from the raw-state router by more than 1.3% except *naive*, at -2.5% ($p = 0.022$, paired over seeds) — and that is the basis-dependent control, not the invariants. The invariant router sits at -0.6% ($p = 0.195$).

We flag one methodological point, because it nearly produced a wrong claim. At three seeds the invariant arm on C-MAPSS averaged 337 ± 71 , apparently far worse than every alternative, and it was tempting to explain the gap. At five seeds it is 296.0 ± 2.8 : the earlier figure was a single divergent initialisation in a small sample. The invariant arm has the largest seed-to-seed spread of any arm we ran, which is itself a finding (Section ??), but the apparent deficit was not real. Paired statistics over at least five seeds are necessary here.

5.2 Is the diffusion assumption even satisfied?

Equation (1) asserts that increments are locally Gaussian with covariance $V(\mathbf{z}) dt$. That is a precondition, not a modelling preference, and it is checkable directly without reference to any downstream accuracy: whiten each held-out increment by its own predicted Cholesky factor, $\mathbf{e} =$

Table 2: Whiteness-increment diagnostics. A correctly specified diffusion gives excess kurtosis 0, a 5σ ratio of $1\times$, small KS distance and whiteness s.d. 1.

system	excess kurtosis	$> 5\sigma$ vs Gaussian	KS (radial)	whiteness s.d.
pmsm	215.5	$8,641\times$	0.740	0.871
cmapss	-1.1	$0\times$	0.170	0.997

$L(\mathbf{z})^{-1}(\Delta\mathbf{z} - \mathbf{b}(\mathbf{z})dt)/\sqrt{dt}$, and ask whether e is standard normal.

The answer inverts the expectation. We anticipated that the simulated benchmark with six discrete flight regimes would violate continuity of paths and that the real continuously-sampled bench would not. The opposite holds. C-MAPSS whitens almost perfectly (excess kurtosis -1.06 , whiteness s.d. 0.997, no increment beyond 5σ), while the real PMSM bench has excess kurtosis 215, a whiteness s.d. of 0.87, and $8,641\times$ the Gaussian rate of 5σ events.

The interpretation is that real machine telemetry contains genuine jumps — commanded set-point changes, load steps, contactor events — that a diffusion cannot represent. The fitted V must absorb them as if they were local noise, which over-shrinks ordinary increments (hence s.d. < 1) while leaving rare ones enormous, and makes $g = V^{-1}$ describe a geometry the machine does not have. We regard this check as the most practically useful output of the work: it is cheap, it is independent of any downstream model, and it tells a practitioner whether the construction is applicable before any of it is built.

5.3 Regime discovery

5.4 Transfer under recalibration

6 Discussion

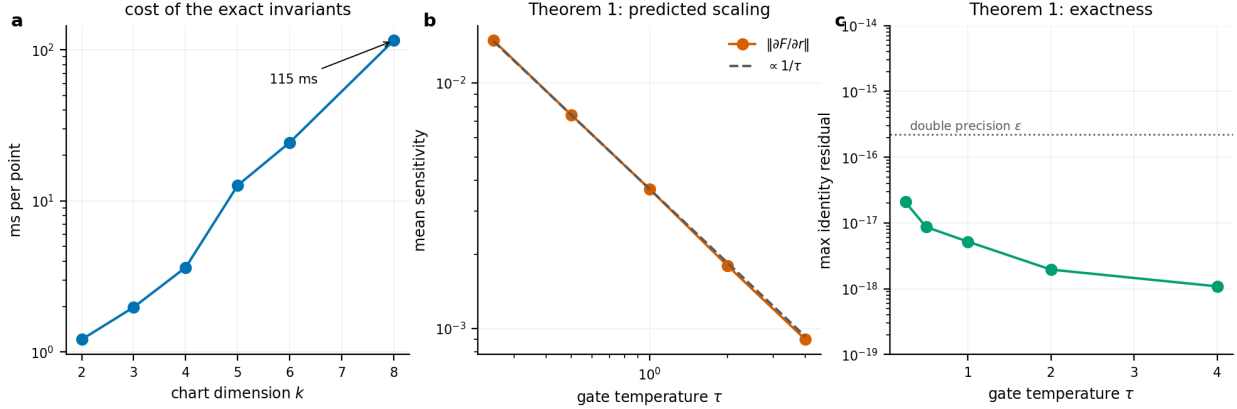
The construction we set out to test is mathematically clean and, in its central claim, correct: the routing coordinates are exact invariants, the metric is the canonical one for a diffusion rather than an arbitrary choice, and the switching behaviour of the assembly obeys an exact identity that yields a real design rule. None of that translated into better in-distribution predictions on any of five systems.

We think the honest reading is threefold. First, the information a geometric router extracts is largely already available to a router that reads the state directly, because the geometry is itself a deterministic function of that state; invariance changes *how* the information is packaged, not how much there is. Second, the coordinate that carries the most physical meaning, the curvature, is the one finite telemetry estimates worst, and its instability propagates into the routing decision — the invariant-only arm has by far the largest seed-to-seed variance of any arm we ran. Third, the promised advantage is conditional: invariance only pays when the coordinate change is severe enough that a raw-state router actually breaks, and mild recalibrations are handled perfectly well by a per-channel quantile match.

What survives is narrower and, we think, still useful: a correct formulation that unifies curvature and volatility into one object at a hundredth of the cost; an exact switching identity with a design consequence; a Pe statistic that is cheap, robustly estimable and exactly invariant, and that is a reasonable regime descriptor in its own right; and a clear, testable precondition — continuity of paths — with a demonstration of what happens when it is violated. The negative accuracy result is itself worth recording, because the intuition behind geometric routing is strong enough that it will be proposed again.

Table 3: The five systems.

system	kind	samples	state d	coupled physics
PMSM bench	real	1,330,816	8	electromagnetic–thermal–mechanical
C-MAPSS FD004	simulated	61,249	17	six flight regimes, two fault modes
UR5e	sim + LPTN	431,520	32	rigid body–contact–thermal
Panda	sim + LPTN	—	37	rigid body–contact–thermal
iiwa14	sim + LPTN	—	37	rigid body–contact–thermal

**Figure 1: Validation.** (a) Cost of the exact invariants against chart dimension; geometry is affordable only on a low-dimensional chart. (b) The $1/\tau$ scaling of routing sensitivity predicted by Theorem 1. (c) Residual of the identity (5), at double-precision epsilon.

The main limitations are that our expert banks are homogeneous, that we do not explore sparse or top- k routing where the switching identity would bite harder, and that curvature estimation might be rescued by stronger regularisation or by estimators that do not differentiate a fitted field twice.

Data availability

All datasets are public. The PMSM electric motor temperature bench and the NASA C-MAPSS turbofan data are on Kaggle ([wkirgsn/electric-motor-temperature](#), [behrad3d/nasa-cmaps](#)). Robot models are from MuJoCo Menagerie. Generated robot telemetry and all result artefacts are produced by the released code.

Code availability

Code to reproduce every figure, table and validation check accompanies this manuscript.

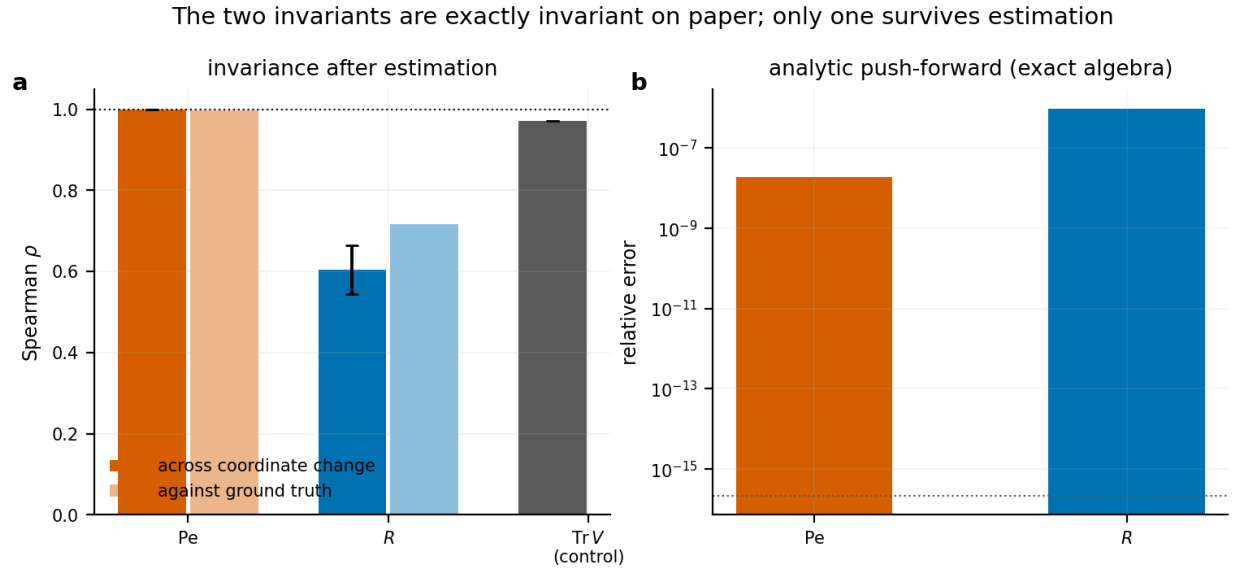


Figure 2: Invariance survives estimation for Pe but not for R . (a) Spearman agreement of each coordinate across a refit in relabelled coordinates, and against ground truth. (b) The same quantities computed analytically, where both are exact.

UR5E episode 3: the coupled subsystems, and the regime they put the machine in

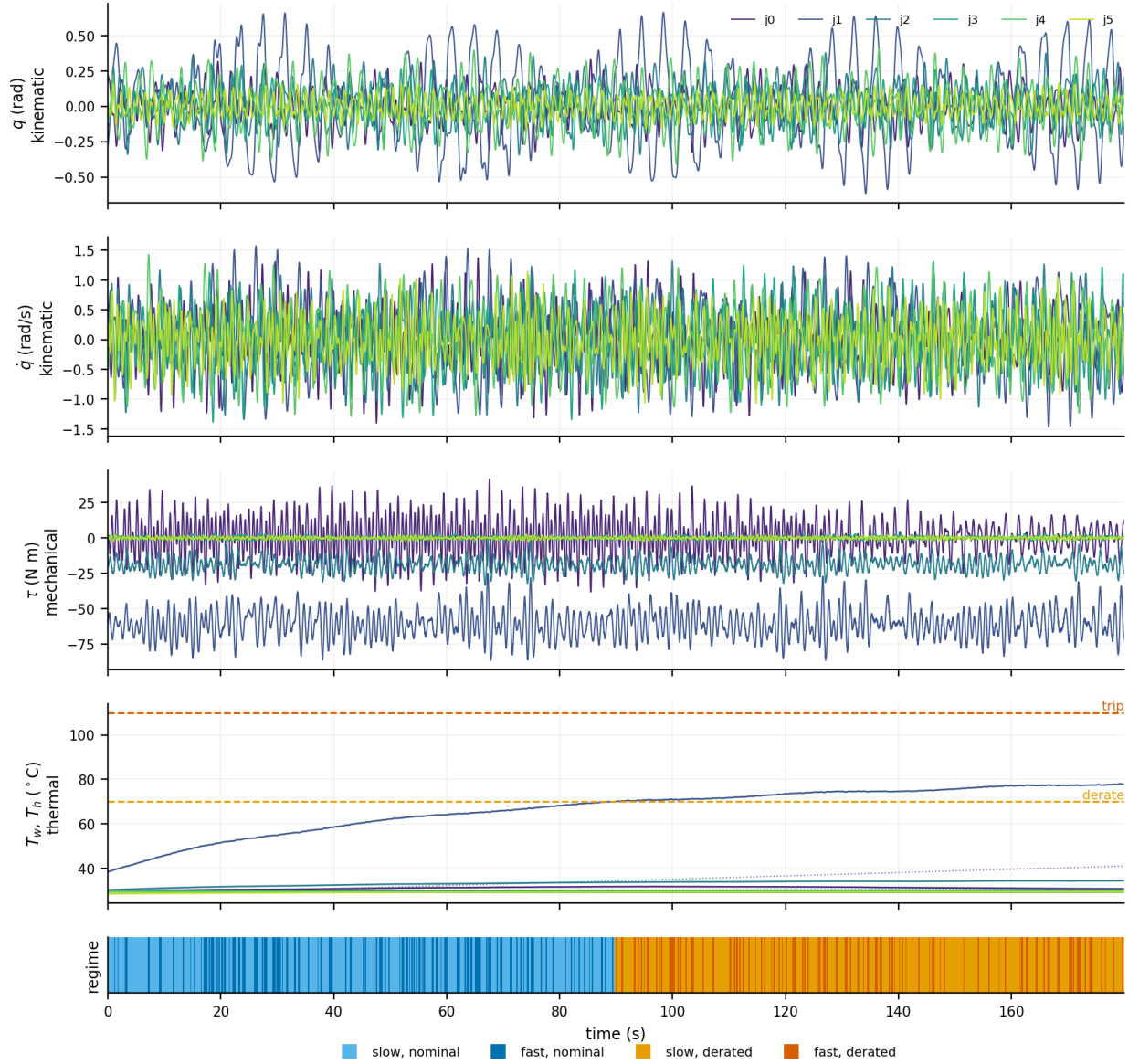


Figure 3: Coupled subsystems on the UR5e. Kinematic and mechanical channels oscillate at ~ 1 Hz while joint 1's winding heats over minutes, crossing the derating knee at $t \approx 90$ s and flipping the ground-truth regime. The three-decade time-scale separation is the multi-physics coupling.

PMSM: the operating manifold in chart coordinates

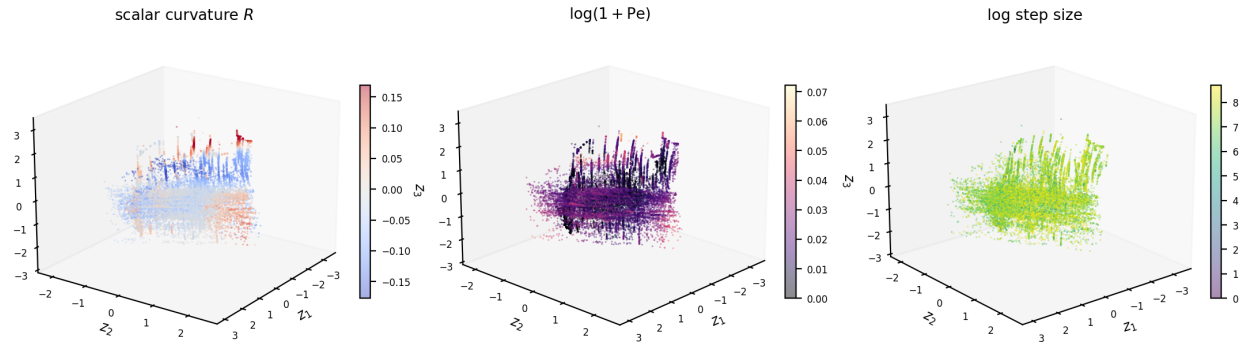


Figure 4: The operating manifold in chart coordinates, coloured by scalar curvature, by $\log(1 + Pe)$ and by step size.